Copernicus Global Land Operations – Lot I

"Vegetation and Energy"

Framework Service Contract N° 945120 – IPR – 2023 (JRC)

PRODUCT USER MANUAL

10-DAY LAND SURFACE TEMPERATURE

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List of Acronyms

AD	Applicable Document
ABI	Advanced Baseline Imager
AHI	Advanced Himawari Imager
ATBD	Algorithm Theoretical Basis Document
CEOS	Committee on Earth Observation Satellites
CF	NetCDF Climate and Forecast (CF) metadata conventions
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud optimized geotiff
ECV	Essential Climate Variable
EPSG	European Petroleum Survey Group
ESA	European Space Agency
EU	European Union
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
DC	Daily Cycle
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEO	Geostationary satellites
GOES	Geostationary Operational Environmental Satellite
HMWR	Himawari
INSPIRE	Infrastructure for Spatial Information in the European Community
IODC	Indian Ocean Data Coverage
IPMA	Instituto Português do Mar e da Atmosfera
JRC	Joint Research Center
LPV	Land Product Validation
LST	Land Surface Temperature
MAE	Mean Absolute Error
MODIS	Moderate Resolution Imaging Spectroradiometer
MSG	Meteosat Second Generation
NetCDF	Network Common Data Form
PQAD	Product Quality Assurance Document
PUM	Product User Manual
RMSE	Root Mean Square Error
SAF	Satellite Application Facility
SEVIRI	Spinning Enhanced Visible Infra-Red Imager
SM	Soil Moisture

TCI	Thermal Conditions Index
TIR	Thermal Infrared
UTC	Coordinated Universal Time
VR	Validation Report
WGS	World Geodetic System

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series. The Product User Manual (PUM) is dedicated to CLMS users giving an overview of the product characteristics, in terms of algorithm, technical properties, and main validation results.

The present document describes the 10-day synthesis of global Land Surface Temperature (10-day LST) product.

The aim of 10-day LST product is to provide a complete overview of the LST daily cycle over each 10-day compositing for every image pixel. The composites will contain the maximum, median and minimum LST values observed per time slot (i.e., 00, 01, …, 23 UTC) during the 10-day period. Those composites can be used to retrieve a synthesis of LST conditions and to derive a Thermal Condition Index (TCI), an index which characterizes the pixel temperature within its expected climatological range. Therefore, a 10-daily TCI is also provided together with the 10-day LST product to ease the use of such information in e. g. agrometeorological applications.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The aim of 10-day LST product is to provide a complete overview of the LST daily cycle over each 10-day compositing for every image pixel. The composites contain the maximum, median and minimum LST values observed per time slot (i.e., 00, 01, …, 23 UTC) during the 10-day period. Those composites can be used to retrieve a synthesis of LST conditions and to derive a Thermal Condition Index (TCI), an index which characterizes the pixel temperature within its expected maximum range. Therefore a 10-day TCI is also provided by the 10-day LST product to ease the use of such information in e.g., agrometeorological applications.

The TCI calculation requires a climatology of LST values close to local noon per pixel, which was specifically estimated for this purpose using all processed data between 2018-2025.

The 10-day LST accuracy is, at least, that of the nominal hourly product, i.e., root mean square differences with respect to 10-day composites of in situ temperature should be within 2 K. The quality assessment of the 10-day LST composites is used as an indication of the TCI validation.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 gives a description of the retrieval algorithm
- Chapter 3 describes the main characteristics of the product
- Chapter 4 shows the main results on validation.

The requirements are recalled in the Annex.

1.3 RELATED DOCUMENTS

Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document for Vegetation and Energy products of the Copernicus Global Land Service
CGLOPS1_ATBD_LST-V3.0	Algorithm Theoretical Basis Document for Land Surface Temperature Version 3.0
CGLOPS1_ATBD_LST10-V3.0	Algorithm Theoretical Basis Document for 10-days Land Surface Temperature Version 3.0
CGLOPS1_PUM_LST-V3.0	Product User Manual for Land Surface Temperature Version 3.0
CGLOPS_VR_LST-LST10-V3.0	Validation Report of the LST and LST10 products Version 2.1

Available at https://land.copernicus.eu/en/technical-library

External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs requirements.

Available Available at https://library.wmo.int/

2 ALGORITHM

The hourly maps of global CLMS LST are used to estimate the main parameters characterizing the daily cycle of LST for each pixel – the maximum, median and minimum temperatures – as well as the fraction of valid pixels used in the calculation. These parameters are estimated for each of the 10 days (dekad). Statistics for the 10-day period are then derived and compared to a multi-year climatology to produce TCI.

2.1 RETRIEVAL METHODOLOGY

The LST10 product is generated every 10 days (at the 1st, 11th, 21st days of each month), with a timeliness up to 2 days, through the following steps (Figure 1):

Step 1: Gather the CLMS LSTv3 product (see CGLOPS1_PUM_LST-V3.0) of all timeslots of the dekad - corresponding to 10 days, except for the last dekad of the month where the period ranges between 8 and 11 days depending on the number of days of the month.

Step 2: Remove positive and negative outliers from the clear sky LST observations keeping only valid LST values.

Step 3: Calculate the decadal statistical parameters (maximum, median, minimum) of the LST product per pixel.

Step 1 to Step 3 are repeated for all the possible timeslots (00, 01, …, 23UTC) to generate a set of 24 maps of statistical parameters obtained for the specific timeslot during the compositing period. The statistical parameters include the maximum, median and minimum values of LST for the whole dekad. In addition, a map with the fraction of valid observations is also provided.

Step 4: Calculate a synthesis of the LST product over the dekad, regardless the time of the day. The synthesis includes the maximum, median and minimum values of LST. The fraction of valid observations in the period is also provided as an ancillary dataset.

Step 5: Calculate the TCI obtained by comparing the median of the LST values observed during the compositing period around the local daily maximum with the upper and lower values of maximum LST from a multi-year climatology.

The climatological pre-processing of LST10 daily cycle product follows:

- Time of local maximum: For each dekad, the multi-annual daily cycle is computed and the time of maximum is retained.
- For each dekad, the multi-annual maximum and minimum of the LST at the time of the local maximum is computed. Pixels with an average fraction of valid data below 0.3 are masked. They are not considered for the climatological calculations neither for the actual TCI

computation. This guarantees a minimum level of sampling for the climatology computations and TCI.

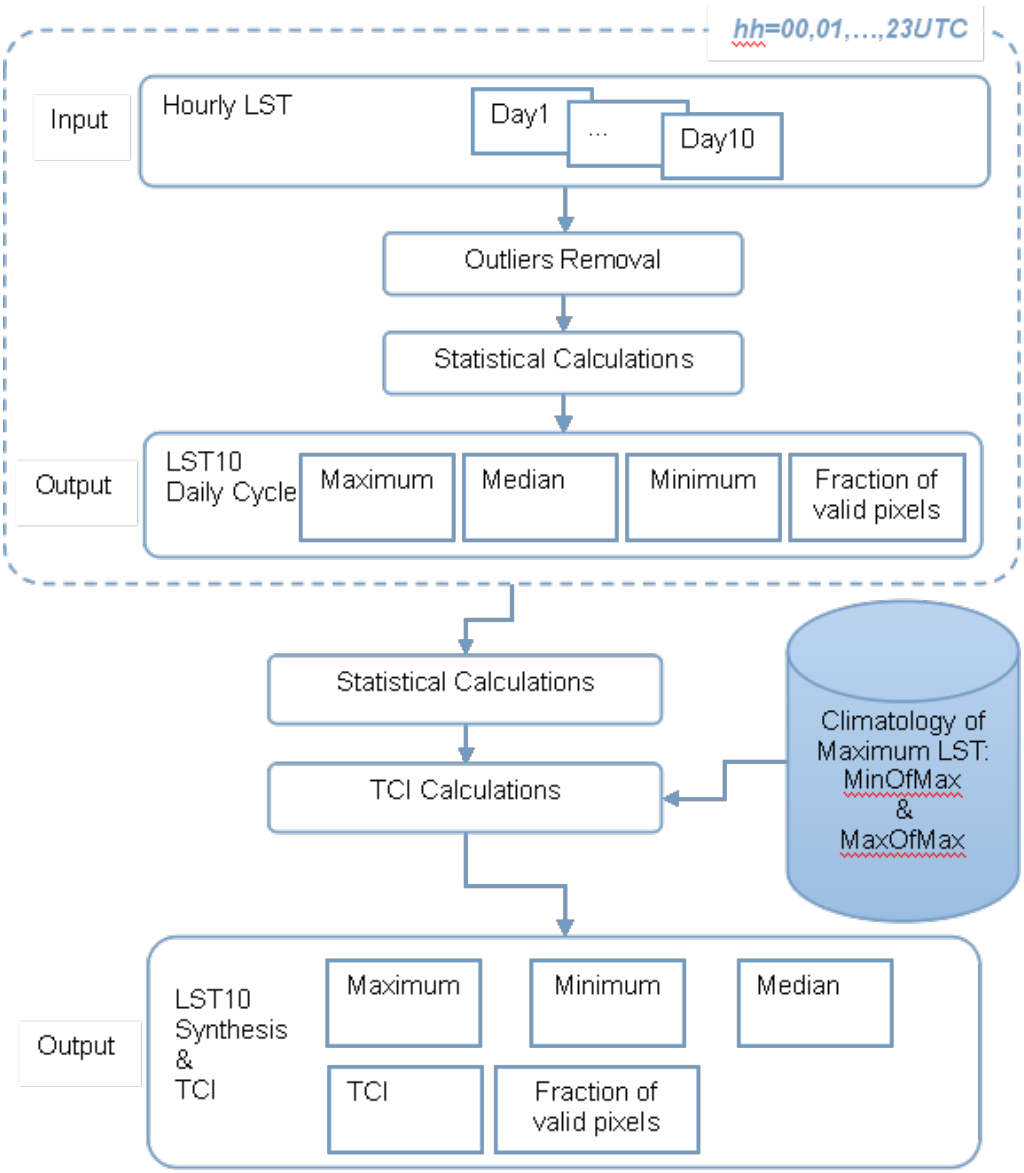


Figure 1: Flow diagram of the generation of 10-day LST products.

2.2 OUTPUT PRODUCTS

LST10 – Daily Cycle

One of the main advantages of retrieving LST from GEO satellites is the ability to characterize the LST daily cycle. A temporal composite of LST is aimed for applications that do not require information on the daily variation of the LST within a relatively short period and/or need an LST daily cycle without gaps due to cloudiness.

The LST10 – Daily Cycle (LST10-DC) is a 10-day composite of LST product for hourly time-slots (00 to 23UTC) and provides maps of the following parameters:

- Maximum, median and minimum of valid LST values observed during the compositing period per timeslot.
- Fraction of valid observations per timeslot in the compositing period used to calculate the statistical parameters.

LST10 – Synthesis & TCI

During the last decades, satellite data has been used to monitor crops and pasture development(e.g., Kogan et al., 2012). It has been demonstrated that vegetation health indices may be derived from moisture and thermal conditions (e.g., Kogan, 2001). Therefore, the 10-day LST composites described above can be used to provide a synthesis of LST conditions of the compositing period and to derive 10-day Thermal Condition Index (TCI) (for more details, see CGLOPS1_ATBD_LST10-V3.0).

2.3 KNOWN LIMITATIONS

It is assumed that the maximum LST value for a specific day occurs at the time of local maximum, computed, for each dekad, from the multi-annual daily cycle, which is a climatological value and could vary from day to day.

Ideally, TCI should be determined with a long climatology of maximum LST. The current data available only covers 8 years (2018-2025).

Since the LST product is only calculated under clear sky conditions, it may happen that in some cloudy areas there are too few valid pixels to compute statistical parameters over the compositing period. However, the number of valid LST values available to calculate those statistics can be used

as an indicator of the robustness of the LST10 product. The use of a threshold of 0.3 for the fraction of valid data is recommended for most applications.

In the 10-daily synthesis product, the maximum and minimum fields should be considered auxiliary fields, as these are more susceptible to sampling uncertainties. The median LST should be treated as the primary synthesis products along with the TCI.

3 PRODUCT DESCRIPTION

Two collections of 10-daily LST products are provided:

- A maximum, minimum and median value composite of the 10-day period, for each hour of a day, thus retaining the main features of the Daily Cycle (LST10-DC).
- Thermal Condition Index, TCI, together with a synthesis of the 10-day period regardless of the observation time (LST10-TCI).

3.1 FILE NAMING

For each 10-day period, the products are named as follows:

- Daily Cycle:

  c_gls_LST10-DC_<YYYYMMDDHHmm>_GLOBE_GEO_V<Major.Minor.Run>.nc

- Synthesis & TCI:

  c_gls_LST10-TCI_<YYYYMMDDHHmm>_GLOBE_GEO_V<Major.Minor.run>.nc

where

- <YYYYMMDDHHmm> gives the temporal location of the file. YYYY, MM and DD denote the year, the month and the day, respectively. The date represents the start of the 10-daily period (dekad) on days 1, 11 and 21 of each month. The HHmm time is set to 0000 by convention.
- GLOBE indicates the global geographical extent of the file;
- GEO indicates that the product is based on geostationary satellite data;
- V<Major.Minor.Run> shows the processing line version used to generate these LST products. The updating of the Major and Minor version and Run number, as well as recommended usage, are described in Table 1.

Table 1: Explanation of the version numbering scheme and recommendations for handling version updates

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same applications unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same applications, but require attention or modest modifications
Run	Fixes to bugs and minor issues.	Later runs are drop-in replacements of former ones.

3.2 FILE FORMAT

The 10-daily LST products are delivered as an internally compressed, hierarchical Network Common Data Form version 4 (netCDF4) data file, with metadata attributes compliant with version 1.6 of the Climate & Forecast conventions (CF V1.6 - http://cfconventions.org/)

3.3 PRODUCT CONTENT

The netCDF4 data file of each 10-daily LST Daily Cycle (LST10-DC) product contains the following data variables:

- **FRAC_VALID_OBS:** Fraction of valid observations as used to calculate the statistical parameters, per hour.
- **MAX, MEDIAN AND MIN:** respectively the maximum, median and minimum LST values observed during the compositing period, per hour.

The netCDF4 data file of 10-daily LST Thermal Condition Index (LST10-TCI) products contains the following data variables:

- **FRAC_VALID_OBS:** Fraction of valid observations as used to calculate the statistical parameters, regardless the time of the day;
- **MAX, MEDIAN AND MIN**: respectively the maximum, median and minimum LST values observed during the compositing period, regardless the time of day;
- **TCI:** Thermal Condition Index for the compositing period, estimated using LST observations around local solar noon;

To convert the digitally stored integer values of the data variables into their physical values, the user may need to multiply the integer values by the Scale Factor and then add the Additional Offset of Table 2. The scale factor and offset are stored in the *scale_factor* and *add_offset* metadata attributes in the netCDF4 files. Modern netCDF tools (e.g., netCDF4 python library) can thus apply the scaling and offset automatically when reading the data variables.

Table 2: Value encoding for the data variables

Variable	Physical values	Unit	Max digital value	Scale Factor	Additional Offset	Flagged values
MIN, MAX, MEDIAN	[-203.15, 353.15] K corresponding to -70 to 80°C	K	8000	1/100	273.15	-9999 Missing
FRAC_VALID_OBS	[0, 1]	-	100	1/100	-	-9999 Missing
TCI	[0, 1]	-	10 000	1/10 000	-	-9999 Missing

As per the CF conventions, the netCDF data files furthermore contain three dimensions and coordinated variables (**lat, lon** and **time**) and a variable for the grid mapping or coordinate reference system (**crs**).

The largely self-describing netCDF files contain a number of metadata attributes:

- on the file-level (Table 3), with example values provided from an LST10-TCI product
- on the level of the data variables, with examples provided for the 'MAX' variable (Table 4) and the 'TCI' variable (Table 5)
- at the level of the coordinate variables for latitude and longitude, that provided as 1-D arrays of 8-byte floating point (Table 6);
- at the level of the coordinate variable for time dimension, that holds one value for the reference or nominal time slot (Table 7)
- at the level of the grid mapping variable (Table 8).

Table 3: File-level metadata attributes of LST10-TCI products.

Attribute	Description	Possible Values
Conventions	Version of the CF-Conventions used	CF-1.6
algorithm_version	Version of the algorithms used to produce the global product.	MSG3-LST_8.0.2,HIMAWARI9-LST_4.2.0,GOES19-LST_4.2.0,MSG2-LST_7.15.2,LST_GLOBE_V3.0
contacts	Contact point(s) for questions or feedback on the LST products.	https://land.copernicus.eu/en/contact-service-helpdesk
copyright	Information on copyright	Copernicus Service Information 2025
history	Provides an audit trail for modifications to the original data netCDF file.	2026-03-23T13:17:53Z: Product generation
Identifier	Unique identifier for the product.	urn:cgls:global:lst10-tci_v3_0.025degree:LST10-TCI_202603110000_GLOBE_GEO_V3.0.1
institution	Specifies the name of the institution that produced the product	IPMA
long_name	Extended product name	10-day Synthesis Land Surface Temperature
orbit_type	Orbit type of the satellites	GEO
parent_identifier	Identifier of the metadata series for the product in Copernicus Global Land metadata catalogue.	urn:cgls:global:lst_V3_0.025degree:LST_202603110000_GLOBE_GEO_V3.0.1
platform	Names of the families (generations) of orbiting platforms	METEOSAT9,METEOSAT10,GOES19,HIMAWARI9,
processing_level	Product processing level	L4
processing_mode	Processing mode used when generating the product, with values Near Real Time, Offline, Reprocessing	Near Real Time
product_version	Version of the product	V3.0.1
references	Published or web-based references with further information about the product.	https://land.copernicus.eu/en/products/temperature-and-reflectance?tab=land_surface_temperature
sensor	Names of the sensors used	SEVIRI,SEVIRI,ABI,AHI,
source	The method of production (e.g. model, observations) of the original data on which the LST product is based.	Derived from EO satellite imagery

Attribute	Description	Possible Values
time_coverage_end	End date and time in UTC of the observation window used, formatted as per ISO 8601.	2026-03-26T07:15:00Z
time_coverage_start	Start date and time in UTC of the observation window used for the product, formatted as per ISO 8601.	2026-03-26T07:00:00Z
title	A brief description of the file content	10-day Synthesis Land Surface Temperature: GLOBE 2026-03-11T00:00:00Z

Table 4: Metadata attributes for the data variable MAX

Attribute	Description	Example value
_FillValue	Value used to represent missing or undefined data	-9999
add_offset	Offset to be added to the variable's contents in order to obtain the real (physical) values	273.15
ancillary_variables	Identification of other closely related variables in the same file	FRAC_VALID_OBS
cell_methods	Methods used to derive the data for each grid cell	area:mean where land
comment	Miscellaneous information about the data or methods used to produce it.	Maximum LST values observed during the compositing period.
coverage_content_type	The coverage content type is an ISO 19115 code list describing the general content type of the resource.	physicalMeasurement
grid_mapping	Reference to the Coordinate Reference System variable	crs
long_name	A descriptive name that indicates a variable's content.	Maximum Land Surface Temperature
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real (physical) values	0.01
units	Units of the variable's content	K
valid_range	Smallest and largest digital values for the variable	-7000, 8000

Table 5: Metadata attributes for the data variable TCI

Attribute	Description	Example value
_FillValue	Value used to represent missing or undefined data	-8000
cell_methods	Methods used to derive the data for each grid cell	area:mean
comment	Miscellaneous information about the data or methods used to produce it.	Thermal Conditions Index for the compositing period estimated using median LST observations at the hour of local maximum.
coverage_content_type	The coverage content type is an ISO 19115 code list describing the general content type of the resource.	physicalMeasurement
grid_mapping	Reference to the Coordinate Reference System variable	crs
long_name	A descriptive name that indicates a variable's content.	Thermal Condition Index
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real (physical) values	0.0001
valid_range	Smallest and largest digital values for the variable	0, 10000

Table 6: Description of netCDF attributes for latitude and longitude coordinate variables

Attribute	Description	Example
axis	Identifies longitude (X), latitude (Y), vertical (Z), or time (T) axes.	X
long_name	A descriptive name that indicates a variable's content.	longitude
standard_name	A standardised name from the CF Convention's standard name table that describes the variable's content.	longitude
units	Units of the variable's content	degrees_east

Table 7: Description of netCDF attributes for time coordinate variable

Attribute	Description	Example
axis	Identifies latitude, longitude, vertical, or time axes.	T
calendar	Specific calendar assigned to the time variable. "standard" or "gregorian" refers to the default mixed Gregorian/Julian calendar.	standard
long_name	A descriptive name that indicates a variable's content.	time
units	Units of the variable's content, taken from Unidata's udunits library.	days since 2019-06-21 00:00:00

Table 8: Description of netCDF attributes for the grid mapping variable

Attribute	Description	Example
grid_mapping_name	Name used to identify the grid mapping or coordinate reference system	latitude_longitude
inverse_flattening	Used to specify the inverse flattening (1/f) of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	298.257223563
longitude_of_prime_meridian	Specifies the longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum	0.0
semi_major_axis	Specifies the length, in metres, of the semi-major axis of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	6378137.0

3.4 PRODUCT CHARACTERISTICS

Projection & Grid Information

The product is displayed in a regular latitude/longitude grid with the ellipsoid WGS 1984 (Terrestrial radius=6378km). This coordinate reference system corresponds to EPSG code 4326. Following CF conventions, this coordinate reference system information is contained within the netCDF4 product file, in the grid_mapping variable called 'crs'. In addition, the coordinate variables for latitude and longitude (1-D arrays) and time (holding 1 value) define the coordinates of each grid cell.

The product resolution is 0.025° (approximately 3km at the equator) and is provided for all land surface pixels from longitude 180°E to 180°W and latitude 70°N to 70°S, resulting in a grid of 14400

columns and 5601 lines. Figure 2 shows the pixel average resolution as function of latitude for the geostationary platforms used and for the version 2 and version 3 LST product.

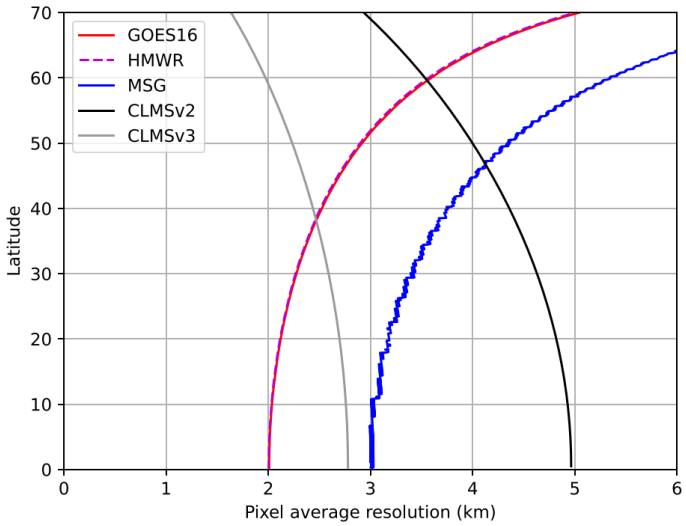


Figure 2 : Pixel average resolution as function of latitude for GOES16 (red), HMWR (dashed magenta), MSG (blue), CLMS LST previous version 2 (black) and CLMS LST version 3 (grey). For the Geostationary, it is assumed the central longitude of each platform.

Temporal Information

The 10-day LST composite is retrieved from the LST instantaneous fields that are estimated every hour (00, 01, 02, 03, …, 23UTC) using representative geostationary (GEO) satellite observations.

The 10-day periods start on days 1, 11 and 21 of each month and end on days 10, 20 and at the end of month respectively. The last period of each month thus consists of between 8 and 11 days. The pixel value can be acquired on any of the days during the respective 10-day period.

3.5 Data Policies

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

3.6 ACCESS AND CONTACTS

The 10-daily LST Daily Cycle and TCI products are available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/ with contact information (helpdesk) on https://land.copernicus.eu/en/contact-service-helpdesk.

4 VALIDATION

The overall quality of the LST10 V3.0 product has been assessed in the validation report [CGLOPS_VR_LST-LST10-V3.0]. A summary is presented below.

An extensive quality assessment was carried out to evaluate the performance of the CLMS LST V3.0 hourly and 10-daily products over the study period from January 2018 to December 2025. The analyses include direct comparisons with in-situ observations from the Copernicus Ground-Based Observations for Validation (GBOV) service and performance assessments using other independent global LST datasets.

Results in the Validation Report [CGLOPS_VR_LST-LST10-V3.0] show that V3.0 performs consistently well with an overall bias of -0.54 K and root mean square difference of 3.20 K when considering the LST10 daily cycle median. Case studies of heatwaves in Europe and southeastern South America illustrate that median LST10-Syn anomalies, combined with the Thermal Condition Index (TCI), effectively capture the spatial distribution and evolution of thermal stress, highlighting the added value of the 10-daily products for monitoring applications.

5 PRODUCT USAGE

5.1 TYPICAL USE

The CLMS LST product provides global, high-frequency observations of land surface temperature derived from a constellation of geostationary satellites, enabling consistent monitoring of surface thermal conditions.

The product is distributed as gridded NetCDF files that can be visualized using standard geospatial software. LST values are obtained by applying the provided scale factor and offset; while missing or low-quality observations (e.g. due to cloud contamination) are identified through dedicated quality flags. Each file represents LST conditions at given reference time, with additional information on acquisition timing and uncertainty provided through ancillary variables.

Standard applications of the product consist mainly in visualization and spatio-temporal analysis. LST fields can be displayed as maps or time-series to identify spatial patterns, seasonal variability, and diurnal temperature variations.

5.2 ADVANCED USAGE

Land surface processes and agriculture

LST is an essential component of the surface energy budget, controlling surface-emitted longwave radiation and enabling estimates of sensible and latent heat fluxes (Trigo et al., 2008). Its response to surface moisture conditions makes it a key variable for land-atmosphere interaction studies.

Under water-limited conditions, LST variations reflect soil moisture (SM) deficits, with diurnal temperature ranges strongly linked to SM variability (Trigo et al., 2015; Anderson et al., 2007). This relationship supports evapotranspiration and SM retrieval methods based on thermal observations (e.g., Li et al., 2021; Ghahremanloo et al., 2019; Tang et al., 2011).

In agricultural applications, geostationary LST captures the diurnal thermal dynamics associated with vegetation water stress, supporting irrigation management and agricultural drought monitoring (e.g., Muro et al., 2018), as well as the development of LST-based drought indices using instantaneous or time-series approaches (e.g., Rhee et al., 2010; Moran et al., 1994).

Hydrology and water cycle

LST is widely used in water resource management and hydrological applications due to its sensitivity to surface energy partitioning and land–atmosphere coupling processes (Trigo et al., 2015; Anderson et al., 2012). It provides indirect information on surface moisture conditions and hydrological variability, supporting the analysis of the surface water balance and drought-related anomalies (Stoyanova et al., 2019). In addition, atmospheric water vapour content is a key factor affecting LST

retrieval accuracy (Freitas et al., 2010). These characteristics support the use of LST in hydrological monitoring and the characterization of land surface water stress at regional and global scales.

Urban Climate

The surface urban heat island refers to the phenomenon in which urban land surfaces exhibit higher temperatures than surrounding rural areas, and is a key indicator of the urban thermal environment (Oke, 1982). Given its suitability for capturing urban heat island diurnal cycles and intra-day variability across cities, urban climate has been extensively studied using LST datasets derived from different satellite platforms and diverse methodologies (e.g., Morgese et al., 2025; Gorai et al., 2024; Naserikia et al., 2023; Elmes et al., 2020; Zhao et al., 2014; Cui & Foy, 2012), providing insight into the temporal evolution and structural characteristics of urban thermal environments.

Thermal anomaly monitoring

LST is suitable for detecting heatwaves and other extreme thermal events by identifying deviations from expected spatial or temporal temperature patterns (e.g., Kim et al., 2025; Martins et al., 2024). Thermal anomalies correspond to positive or negative departures from a reference LST state and are commonly identified using statistical thresholds applied to spatial, temporal, or combined climatological baselines (Li et al., 2023). These approaches enable the detection of anomalous surface temperature conditions, supporting the identification of localized events as well as large-scale, persistent thermal extremes such as heatwaves, relevant for climate and environmental monitoring.

Climate studies

LST is increasingly applied in climate studies to monitor variability and support the derivation of climate change indices (Li et al., 2023). Small shifts in mean surface temperature can lead to disproportionate changes in extremes, making thermal extremes particularly relevant for climate studies (e.g., Reiners et al., 2023; Caloiero, 2017). In this context, LST observations provide valuable complementary information to air temperature datasets, especially in regions with limited in-situ coverage or coarse-resolution reanalysis products.

Satellite–derived LST can be used to estimate climate indices related to temperature extremes, including intensity (e.g. maximum and minimum LST) and range metrics such as the diurnal temperature range. These indicators support the analysis of climate variability, long-term trends, and changes in extreme thermal conditions from regional to global scales (e.g., Alexander et al., 2006).

Model-related applications

LST is widely used in model evaluation (e.g., Ghent et al., 2024; Martin et al., 2019) due to its spatial coverage and sensitivity to land–atmosphere processes, enabling validation and intercomparison of land surface and climate models, particularly in the assessment of surface energy balance and temperature-related processes. Evaluation typically relies on comparisons with in-situ observations using standard performance metrics (e.g., bias, RMSE, MAE, coefficient of determination).

In addition, satellite–derived LST is increasingly assimilated into numerical models to constrain surface states and improve the representation of land–atmosphere interactions. It also contributes to reanalysis systems by providing observational constraints that enhance the consistency and accuracy of long-term climate datasets.

Beyond evaluation and assimilation, LST is also used in data-driven modelling approaches, including machine learning methods, to represent and predict surface temperature dynamics (e.g., Hoang et al., 2025; Pandle et al., 2024). These approaches support applications such as urban climate analysis and environmental monitoring.

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7 ANNEX: REQUIREMENTS

According to the applicable document [AD1], specific technical details and requirements are described below (Table 9).

7.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

Table 9: CGLOPS LST product specifications.

LST – LAND SURFACE TEMPERATURE	
The Land Surface Temperature (LST) is the radiative skin temperature of the land surface, as measured in the direction of the remote sensor [K].	
Geometric Properties:	
Spatial resolution	5 km
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global
Time span	Hourly and 10-day period
Timeliness	Hourly: within 4 hours; within 2 days (optimally 1 day) after the end of 10-day period
Uncertainty (2-sigma)	Threshold: 2 K, Goal: 1 K
Stability (per decade)	Threshold: 0.3 K, Goal: 0.1 K

7.2 FURTHER REQUIREMENTS

7.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

7.2.2 Data structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, and conversion to physical values formulas.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product;
- The per-pixel date of the individual measurement or the start-end dates of the period covered;
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result;

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

7.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format, such as COG or ZAR, format can be proposed for production or requested by the contracting authority during the Framework.

7.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications (Table 9) follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate, the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL datasets and/or Ground-Based Observations for Validation (GBOV, https://land.copernicus.eu/en/products/GBOV) of Copernicus Global Land component when biophysical parameters are available.

7.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu;

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary datasets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

7.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic